



Department of Computer Science and Engineering

DBMS Unit 2 Question Bank

	COGNITIVE LEVEL : EASY
1	Inventory Update Relational Schema: PRODUCT(ProductID PK, ProductName, Stock) Q) A retail store stores PRODUCT(ProductID, ProductName, Stock). A shipment adds 25 units to product 104. Write the UPDATE statement that increases its stock without changing any other product.
2	Customer Registration Relational Schema: CUSTOMER(CustomerID PK, Name, Email, City) Q) An online marketplace stores CUSTOMER(CustomerID, Name, Email, City). Write an INSERT statement to register a new customer with a given ID, name, email and city.
3	Order Status Modification Relational Schema: ORDER_TABLE(OrderID PK, CustomerID FK, Status) Q) A food-delivery company stores ORDER_TABLE(OrderID, CustomerID, Status). A cancelled order must be marked as 'Cancelled'. Write the UPDATE statement for a specified order.
4	Expired Coupon Removal Relational Schema: COUPON(CouponID PK, Code, ExpiryStatus) Q) An e-commerce company stores COUPON(CouponID, Code, ExpiryStatus). Write a DELETE statement that removes coupons whose ExpiryStatus is 'Expired'.
5	Hospital Patient Listing Relational Schema: PATIENT(PatientID PK, Name, Department, Age) Q) A hospital stores PATIENT(PatientID, Name, Department, Age). Display the names and departments of patients older than 60, ordered from oldest to youngest.
6	Unique Delivery Cities Relational Schema: DELIVERY(DeliveryID PK, CustomerID FK, City)

	Q) A courier company stores DELIVERY(DeliveryID, CustomerID, City). Display the distinct cities to which deliveries have been made.
7	Unpaid Invoices Relational Schema: INVOICE(InvoiceID PK, StudentID FK, Amount, PaymentStatus) Q) A college stores INVOICE(InvoiceID, StudentID, Amount, PaymentStatus). Display invoices whose PaymentStatus is NULL because payment information has not yet been recorded.
8	Monthly Sales Total Relational Schema: SALE(SaleID PK, ProductID FK, Quantity, Amount) Q) A store stores SALE(SaleID, ProductID, Quantity, Amount). Find the total sales amount generated by each product.
9	Department Employee Count Relational Schema: EMPLOYEE(EmpID PK, Name, DepartmentID FK) Q) A company stores EMPLOYEE(EmpID, Name, DepartmentID). Find the number of employees working in each department.
10	Exam Result Summary Relational Schema: EXAM_RESULT(StudentID PK/FK, SubjectID PK/FK, Score) Q) A university stores EXAM_RESULT(StudentID, SubjectID, Score). For a selected subject, find the highest, lowest and average score.
11	Metro Stations Relational Schema: TRAIN(TrainID PK, TrainName); STATION(StationID PK, StationName); TRAIN_STATION(TrainID FK, StationID FK) Q) A metro database stores TRAIN(TrainID, TrainName), STATION(StationID, StationName), and TRAIN_STATION(TrainID, StationID). Display the stations served by a selected train.
12	University Clubs Relational Schema: CLUB(ClubID PK, ClubName); CLUB_MEMBER(StudentID FK, ClubID FK); STUDENT(StudentID PK, StudentName) Q) A university stores CLUB(ClubID, ClubName) and CLUB_MEMBER(StudentID, ClubID). Display the names of students belonging to a selected club.
13	Hospital Departments Relational Schema: DOCTOR(DoctorID PK, Name, DepartmentID FK); DEPARTMENT(DepartmentID PK, DepartmentName) Q) A hospital stores DOCTOR(DoctorID, Name, DepartmentID) and DEPARTMENT(DepartmentID, DepartmentName). Display each doctor with the department name.

14	Airline Destinations Relational Schema: FLIGHT(FlightID PK, DestinationID FK); AIRPORT(AirportID PK, City) Q) An airline stores FLIGHT(FlightID, DestinationID) and AIRPORT(AirportID, City). Display each flight with its destination city.
15	Restaurant Menu Relational Schema: DISH(DishID PK, Name, CategoryID FK); CATEGORY(CategoryID PK, CategoryName) Q) A restaurant stores DISH(DishID, Name, CategoryID) and CATEGORY(CategoryID, CategoryName). Display each dish with its category.
16	Courier Tracking Relational Schema: PACKAGE(PackageID PK, CustomerID FK, Status); CUSTOMER(CustomerID PK, Name) Q) A courier stores PACKAGE(PackageID, CustomerID, Status) and CUSTOMER(CustomerID, Name). Display customer names with the current status of their packages.
17	Bank Branches Relational Schema: ACCOUNT(AccountID PK, CustomerID FK, BranchID FK); BRANCH(BranchID PK, BranchName) Q) A bank stores ACCOUNT(AccountID, CustomerID, BranchID) and BRANCH(BranchID, BranchName). Display every account with its branch.
18	Library Member Sets Relational Schema: ACTIVE_MEMBER(MemberID PK); PREMIUM_MEMBER(MemberID PK) Q) A library stores ACTIVE_MEMBER(MemberID) and PREMIUM_MEMBER(MemberID). Produce the list of members who belong to both sets.
19	Railway Passenger Sets Relational Schema: WEEKDAY_PASSENGER(PassengerID PK); WEEKEND_PASSENGER(PassengerID PK) Q) A railway stores WEEKDAY_PASSENGER(PassengerID) and WEEKEND_PASSENGER(PassengerID). Produce the passengers who travel on weekdays but not on weekends.
20	Streaming Subscriber Sets Relational Schema: MONTHLY_SUBSCRIBER(UserID PK); ANNUAL_SUBSCRIBER(UserID PK) Q) A streaming service stores MONTHLY_SUBSCRIBER(UserID) and ANNUAL_SUBSCRIBER(UserID). Produce the list of users subscribed to either plan without duplicates.

21	Sports Performance Relational Schema: PLAYER(PlayerID PK, Name, Score) Q) A sports academy stores PLAYER(PlayerID, Name, Score). Find players whose score is above the overall average score using a subquery.
22	Job Market Relational Schema: JOB(JobID PK, Company, Role, Salary) Q) A recruitment portal stores JOB(JobID, Company, Role, Salary). Find all jobs offering the maximum salary, including ties.
23	Bookstore Pricing Relational Schema: BOOK(BookID PK, CategoryID FK, Price); CATEGORY(CategoryID PK, CategoryName) Q) A bookstore stores BOOK(BookID, CategoryID, Price). Find books priced above the average price of their category using a subquery.
24	Hospital Capacity Relational Schema: WARD(WardID PK, Capacity); BED(BedID PK, WardID FK, Status) Q) A hospital stores WARD(WardID, Capacity) and BED(BedID, WardID, Status). Find wards whose number of occupied beds exceeds half their capacity.
25	University Scholarships Relational Schema: SCHOLARSHIP(StudentID PK/FK, Amount); STUDENT(StudentID PK, DepartmentID FK) Q) A university stores SCHOLARSHIP(StudentID, Amount) and STUDENT(StudentID, DepartmentID). Find students receiving more scholarship than the overall average scholarship.
26	Restaurant Reviews Relational Schema: REVIEW(ReviewID PK, RestaurantID FK, Rating); RESTAURANT(RestaurantID PK, Name) Q) A restaurant platform stores REVIEW(RestaurantID, Rating). Find restaurants whose average rating is higher than the overall average rating.
27	Stock Market Relational Schema: STOCK_PRICE(Symbol PK/FK, TradingDate PK, ClosePrice); STOCK(Symbol PK, CompanyName) Q) A market stores STOCK_PRICE(Symbol, TradingDate, ClosePrice). Find trading days on which a stock closed above its historical average closing price.
28	Airline Flights Relational Schema: FLIGHT(FlightID PK, AirlineID FK, Duration); AIRLINE(AirlineID PK, AirlineName) Q) An airline stores FLIGHT(FlightID, AirlineID, Duration). Find airlines that operate at least one flight longer than the average flight duration across all flights.

29	Museum Valuation Relational Schema: ARTIFACT(ArtifactID PK, PeriodID FK, Value); PERIOD(PeriodID PK, PeriodName) Q) A museum stores ARTIFACT(ArtifactID, PeriodID, Value). Find artifacts valued above the overall average artifact value.
30	Insurance Claims Relational Schema: CLAIM(ClaimID PK, PolicyID FK, ClaimAmount); POLICY(PolicyID PK, PolicyType) Q) An insurer stores CLAIM(PolicyID, ClaimAmount). Find policies having at least one claim larger than the average claim amount.
31	Second Highest Score Q) A university stores EXAM_RESULT(StudentID, SubjectID, Score). For a selected subject, return the second-highest distinct score. If fewer than two distinct scores exist, return NULL.
32	Consecutive Values Q) A monitoring system stores SENSOR_LOG(LogID, Value). Find values that occur at least three times consecutively when rows are ordered by LogID.
33	Department Top Three Salaries Q) A company stores EMPLOYEE(EmpID, DepartmentID, Salary). Return employees whose salary is among the top three distinct salaries within their department.
34	City Temperature Comparison Q) A climate system stores DAILY_TEMP(City, DateRecorded, Temperature). For every city, find dates where temperature increased compared with the previous recorded day for that city.
35	Third Transaction Q) A payment system stores TRANSACTION(UserID, TransactionTime, Amount). For every user who has made at least three transactions, return the time and amount of their third chronological transaction.
	COGNITIVE LEVEL : MEDIUM
36	Rolling Average Q) A food-delivery platform stores DAILY_SALES(RestaurantID, SalesDate, Revenue). Calculate a 7-day rolling average of revenue for each restaurant.
37	Contest Leaderboard Q) A coding platform stores SUBMISSION(HackerID, ChallengeID, Score). Calculate each hacker's total score using the best submission per challenge, then produce the leaderboard in descending score order with ties handled appropriately.

38	Employee Bonus Q) A company stores EMPLOYEE(EmpID, Name, Salary) and BONUS(EmpID, BonusAmount). Find employees whose bonus is below 1000 or who have no bonus record at all.
39	Trip Cancellation Analysis Q) A ride-sharing system stores TRIPS(TripID, ClientID, DriverID, Status, TripDate) and USERS(UserID, Banned). Calculate the cancellation rate for each day considering only trips where neither client nor driver is banned.
40	Department Salary Comparison Q) A company stores EMPLOYEE(EmpID, DepartmentID, Salary). Return every employee together with the average salary of their department and a flag indicating whether the employee earns above, equal to, or below that average.
41	First Login Q) A gaming platform stores GAME_PLAYERS(PlayerID, DeviceID, EventDate). Find each player's first login date.
42	Confirmation Rate Q) A mobile application stores SIGNUPS(UserID, SignupDate) and CONFIRMATIONS(UserID, Time, Action), where Action is confirmed or timeout. Calculate each user's confirmation rate.
43	Product Sales Analysis Q) A retailer stores PRODUCT(ProductID, Name) and SALES(ProductID, SaleDate, Quantity). Find products that were sold only during a specified year and not outside that year.
44	Preferred Delivery Q) A food-delivery platform stores DELIVERY(DeliveryID, CustomerID, OrderDate, PreferredDate). Calculate the percentage of orders delivered on the customer's preferred date.
45	Year-over-Year Growth Q) A company stores MONTHLY_REVENUE(ProductID, Month, Revenue). For every product, calculate the year-over-year percentage change in revenue.
46	Repeated Payment Amounts Relational Schema: PAYMENT(PaymentID PK, AccountID FK, Amount) Q) A payment gateway stores PAYMENT(PaymentID, AccountID, Amount). Find accounts that have made at least two payments of the same amount.
47	Customer Repeat Purchase Relational Schema: ORDER_TABLE(OrderID PK, CustomerID FK, ProductID FK)

	Q) An online store stores ORDER_TABLE(OrderID, CustomerID, ProductID). Find customers who have purchased the same product at least twice.
48	Challenge Leaderboard Q) A coding contest stores SUBMISSION(HackerID, ChallengeID, Score) and CHALLENGE(ChallengeID, Difficulty). Find the competitors who achieved full scores in the largest number of distinct challenges.
49	Student Report Q) A university stores STUDENT(Name, Marks, Grade) and wants a report ordered by grade and marks, but names should be hidden for students below a specified grade threshold. Construct the query using CASE and ORDER BY.
50	Department Attendance Analysis Relational Schema: ATTENDANCE(StudentID PK/FK, CourseID PK/FK, Present) Q) A university stores ATTENDANCE(StudentID, CourseID, Present). Find courses whose average attendance percentage is above the overall course-average attendance.
51	Library Fine Function Relational Schema: LOAN(LoanID PK, MemberID FK, DueDate, ReturnDate); FINE_RULE(DailyRate) Q) Create a SQL function that accepts days overdue and a daily fine rate and returns the total fine.
52	Taxi Fare Function Relational Schema: TRIP(TripID PK, Distance, BaseFare, RatePerKm) Q) Create a SQL function that calculates taxi fare from distance and a per-kilometre rate, with a fixed base fare.
53	Payroll Procedure Relational Schema: EMPLOYEE(EmpID PK, Name, DepartmentID FK, Salary); DEPARTMENT(DepartmentID PK, DepartmentName) Q) Create a stored procedure that accepts a department ID and returns all employees in that department along with their current salary.
54	Hospital Appointment Procedure Relational Schema: APPOINTMENT(AppointmentID PK, DoctorID FK, AppointmentDate, AppointmentTime); DOCTOR(DoctorID PK, Name) Q) Create a stored procedure that accepts a doctor ID and date and returns all appointments scheduled for that doctor on that date.
55	Currency Conversion Function Relational Schema: EXCHANGE_RATE(FromCurrency PK, ToCurrency PK, Rate) Q) Create a function that converts an amount from one currency to another using a supplied exchange rate.

56	Courier Status Procedure Relational Schema: PACKAGE(PackageID PK, Status, UpdatedAt); PACKAGE_TRACKING(TrackingID PK, PackageID FK, Status, UpdatedAt) Q) Create a stored procedure that accepts a package ID and returns its latest tracking status and timestamp.
57	University Attendance CTE Relational Schema: ATTENDANCE(StudentID FK, ClassDate PK, Present); STUDENT(StudentID PK, Name) Q) Using a CTE, calculate each student's overall attendance percentage and return students below 75%.
58	Electricity Bill Analysis Relational Schema: USAGE(MeterID FK, Month PK, Units); HOUSEHOLD(MeterID PK, HouseholdName, City) Q) Using a CTE, first calculate monthly consumption for each household and then identify households whose annual consumption exceeds the city average.
59	Library Borrowing CTE Relational Schema: BORROW(MemberID FK, BookID FK, BorrowDate); MEMBER(MemberID PK, Name) Q) Use a CTE to calculate total books borrowed by each member and then classify members as Low, Medium or High borrowers using CASE.
60	Hotel Occupancy CTE Relational Schema: ROOM(RoomID PK, HotelID FK); HOTEL_DAY(RoomID FK, Date PK, Occupied) Q) Use a CTE to calculate daily occupied rooms and then identify days where occupancy exceeded 90%.
61	University Department CTE Relational Schema: STUDENT(StudentID PK, DepartmentID FK); DEPARTMENT(DepartmentID PK, Name) Q) A university stores STUDENT(StudentID, DepartmentID) and DEPARTMENT(DepartmentID, Name). Using a CTE, calculate the number of students in each department and display departments with more than 100 students.
62	Retail Sales CTE Relational Schema: SALE(SaleID PK, ProductID FK, Quantity, Amount); PRODUCT(ProductID PK, ProductName) Q) A retailer stores SALE(SaleID, ProductID, Quantity, Amount). Using a CTE, calculate total sales amount for each product and return products whose sales exceed the average product sales.
63	Hospital CTE Relational Schema: APPOINTMENT(AppointmentID PK, DoctorID FK, Status); DOCTOR(DoctorID PK, Name)

	Q) A hospital stores APPOINTMENT(AppointmentID, DoctorID, Status). Using a CTE, calculate completed appointments for each doctor and return doctors with more than 50 completed appointments.
64	Library CTE Relational Schema: BORROW(BorrowID PK, MemberID FK, BookID FK); MEMBER(MemberID PK, Name) Q) A library stores BORROW(BorrowID, MemberID, BookID). Using a CTE, calculate the number of books borrowed by each member and return members who borrowed more than the average member.
65	Bank CTE Relational Schema: ACCOUNT(AccountID PK, CustomerID FK); TRANSACTION(TransactionID PK, AccountID FK, Amount, TransactionType) Q) A bank stores ACCOUNT(AccountID, CustomerID) and TRANSACTION(TransactionID, AccountID, Amount, TransactionType). Using a CTE, calculate each customer's total credited amount and return customers whose total credits exceed ₹5 lakh.
66	Cinema CTE Relational Schema: BOOKING(BookingID PK, ScreenID FK, TicketPrice); SCREEN(ScreenID PK, ScreenName) Q) A cinema stores BOOKING(BookingID, ScreenID, TicketPrice). Using a CTE, calculate total booking revenue for each screen and return screens whose revenue is above the average screen revenue.
67	Food Delivery Growth Relational Schema: ORDER_TABLE(OrderID PK, RestaurantID FK, OrderDate, Amount); RESTAURANT(RestaurantID PK, Name) Q) Use a CTE to calculate each restaurant's monthly revenue and then find restaurants whose revenue increased for two consecutive months.
68	Hospital Workload Relational Schema: APPOINTMENT(AppointmentID PK, DoctorID FK, AppointmentDate); DOCTOR(DoctorID PK, Name) Q) Use a CTE to calculate monthly appointments per doctor and identify doctors whose monthly workload exceeds the hospital-wide monthly average.
69	Transport Delays Relational Schema: TRIP(TripID PK, RouteID FK, ScheduledTime, ActualTime); ROUTE(RouteID PK, RouteName) Q) Use a CTE to calculate average delay per route and then find routes whose average delay is above the city-wide route average.
70	Stock Portfolio Relational Schema: PORTFOLIO(InvestorID FK, Symbol FK, Quantity); INVESTOR(InvestorID PK, Name); STOCK(Symbol PK, CurrentPrice)

	Q) Use a CTE to calculate the total current value of each investor's portfolio and return investors whose value exceeds ₹10 lakh.
	COGNITIVE LEVEL : HARD
71	Electricity Month Comparison Relational Schema: USAGE(HouseholdID FK, UsageMonth PK, Units); HOUSEHOLD(HouseholdID PK, Name) Q) For every household, show each month's electricity usage and the usage in the previous recorded month.
72	Stock Price Movement Relational Schema: STOCK_PRICE(Symbol FK, TradingDate PK, ClosePrice); STOCK(Symbol PK, CompanyName) Q) For each stock, show the current closing price, previous closing price and percentage change.
73	Metro Passenger Ranking Relational Schema: STATION_MONTHLY(City, StationID FK, Month PK, PassengerCount); STATION(StationID PK, StationName, City) Q) Rank metro stations within each city according to monthly passenger count, giving equal ranks to equal counts.
74	Hospital Doctor Ranking Relational Schema: APPOINTMENT(DoctorID FK, AppointmentDate PK, Status); DOCTOR(DoctorID PK, Name) Q) For each month, rank doctors by number of completed appointments and return the top three ranks.
75	Streaming Watch History Relational Schema: VIEW_SESSION(UserID FK, SessionDateTime PK, ContentID FK); USER(UserID PK, Name) Q) For every user, show the date of their current viewing session and the date of their previous viewing session.
76	Warehouse Restocking Relational Schema: STOCK(ProductID FK, WarehouseID FK, RecordedAt PK, Quantity); PRODUCT(ProductID PK); WAREHOUSE(WarehouseID PK) Q) For every product, show the current stock, previous recorded stock and the change in quantity.
77	Airline On-Time Ranking Relational Schema: FLIGHT(AirlineID FK, FlightID PK, Quarter, OnTime); AIRLINE(AirlineID PK, AirlineName) Q) Rank airlines within each quarter according to their on-time arrival percentage.

78	Football Tournament Relational Schema: MATCH(MatchID PK, TeamID FK, MatchDate, Result); TEAM(TeamID PK, TeamName) Q) For each team, calculate points earned after every match using a running total, where win = 3, draw = 1 and loss = 0.
79	Bank Transactions Relational Schema: TRANSACTION(TransactionID PK, AccountID FK, TransactionTime, TransactionType, Amount); ACCOUNT(AccountID PK) Q) For each account, calculate the running account balance from a sequence of credit and debit transactions.
80	Restaurant Revenue Relational Schema: REVENUE(RestaurantID FK, RevenueMonth PK, Revenue); RESTAURANT(RestaurantID PK, Name) Q) For every restaurant, calculate monthly revenue and the percentage difference from the previous month.
81	Support Ticket Search Relational Schema: SUPPORT_TICKET(TicketID PK, CustomerID FK, Subject, Description, CreatedAt) Q) A company has millions of support tickets. Design a full-text search solution for finding tickets related to “payment” and “refund” without scanning every row with LIKE.
82	Digital Library Search Relational Schema: BOOK(BookID PK, Title, Summary, Keywords, AuthorID FK); AUTHOR(AuthorID PK, Name) Q) A digital library stores book titles, summaries and keywords. Design a full-text search query that allows users to search for books related to “distributed systems”.
83	News Archive Relational Schema: ARTICLE(ArticleID PK, Title, Body, PublishedAt, CategoryID FK) Q) A news portal stores millions of articles. Explain how full-text indexing can support searches for articles discussing “climate change” while ranking the most relevant results first.
84	Product Search Relational Schema: PRODUCT(ProductID PK, Name, Description, Features, CategoryID FK) Q) An e-commerce site wants a search box that finds products related to “wireless noise cancelling headphones”. Explain how full-text search differs from LIKE and what fields should be indexed.
85	Legal Document Search Relational Schema: CASE_DOCUMENT(DocumentID PK, CaseID FK, Title, DocumentText); CASE_FILE(CaseID PK, CaseTitle)

	Q) A legal database stores case titles and document text. Design a full-text search approach for finding documents containing “data privacy” and explain how phrase search differs from independent keyword search.
86	Medical Knowledge Base Relational Schema: KNOWLEDGE_ARTICLE(ArticleID PK, Title, Body, CategoryID FK) Q) A hospital knowledge base stores article titles and article bodies. Design a search requirement for finding articles about “drug interaction” and explain why an index is important.
87	Code Repository Search Relational Schema: ISSUE(IssueID PK, RepositoryID FK, Title, Description, CreatedAt); REPOSITORY(RepositoryID PK, Name) Q) A software company stores commit messages and issue descriptions. Design a full-text search that can find issues related to “database connection timeout”.
88	Customer Feedback Search Relational Schema: REVIEW(ReviewID PK, ProductID FK, ReviewText, Rating, CreatedAt); PRODUCT(ProductID PK) Q) A company wants to find reviews mentioning both “delivery” and “damaged”. Design a full-text search condition and explain how relevance ranking could improve the result order.
89	Travel Review Search Relational Schema: HOTEL_REVIEW(ReviewID PK, HotelID FK, ReviewText, Rating); HOTEL(HotelID PK, HotelName) Q) A travel site stores hotel reviews. Design a search that finds reviews related to “airport shuttle” and discuss the difference between natural-language and exact phrase searching.
90	Internal Knowledge Search Relational Schema: TECH_DOCUMENT(DocumentID PK, Title, Body, DepartmentID FK); DEPARTMENT(DepartmentID PK, DepartmentName) Q) A company stores technical documents and wants employees to search for “authentication token expired”. Propose the full-text indexed columns and a suitable query strategy.
91	Bank Customer Risk Relational Schema: CUSTOMER(CustomerID PK, Name); ACCOUNT(AccountID PK, CustomerID FK, Balance); TRANSACTION(TransactionID PK, AccountID FK, Amount) Q) A bank stores CUSTOMER(CustomerID, Name), ACCOUNT(AccountID, CustomerID, Balance) and TRANSACTION(TransactionID, AccountID, Amount). Produce a report of customers whose total account balance is above the average customer balance and who have made at least five transactions.

92	E-Commerce Product Analysis Relational Schema: PRODUCT(ProductID PK, CategoryID FK, Price); ORDER_TABLE(OrderID PK, CustomerID FK, ProductID FK, Quantity) Q) An online store stores PRODUCT(ProductID, CategoryID, Price), ORDER_TABLE(OrderID, CustomerID, ProductID, Quantity). Find categories whose total sales quantity is above the average category sales quantity and rank products within each such category by quantity sold.
93	Hospital Doctor Performance Relational Schema: DOCTOR(DoctorID PK, DepartmentID FK); APPOINTMENT(AppointmentID PK, DoctorID FK, Status); DEPARTMENT(DepartmentID PK, Name) Q) A hospital stores DOCTOR(DoctorID, DepartmentID), APPOINTMENT(AppointmentID, DoctorID, Status) and DEPARTMENT(DepartmentID, Name). Find doctors whose completed appointment count is above the average for their department and rank them within their department.
94	University Course Performance Relational Schema: COURSE(CourseID PK, DepartmentID FK); ENROLLMENT(StudentID FK, CourseID FK, Grade) Q) A university stores COURSE(CourseID, DepartmentID), ENROLLMENT(StudentID, CourseID, Grade). Find courses whose average grade is above the overall course average and rank those courses within their department by average grade.
95	Retail Customer Spending Relational Schema: CUSTOMER(CustomerID PK, Name, City); ORDER_TABLE(OrderID PK, CustomerID FK, Amount) Q) A retailer stores CUSTOMER(CustomerID, Name), ORDER_TABLE(OrderID, CustomerID, Amount). Calculate each customer's total spending and rank customers within each city, returning customers whose spending is above the average spending of customers in their city.
96	Movie Booking Analysis Relational Schema: MOVIE(MovieID PK, Title); SHOW(ShowID PK, MovieID FK, ScreenID FK); BOOKING(BookingID PK, ShowID FK, Seats) Q) A cinema stores MOVIE(MovieID, Title), SHOW(ShowID, MovieID, ScreenID), BOOKING(BookingID, ShowID, Seats). Find movies whose total seats booked are above the average movie booking total and rank movies by seats booked.
97	Employee Salary Analysis Relational Schema: EMPLOYEE(EmpID PK, DepartmentID FK, Salary) Q) A company stores EMPLOYEE(EmpID, DepartmentID, Salary). Find departments whose average salary is above the company-wide average and rank employees in each such department by salary, including ties.
98	Library Borrowing Analysis Relational Schema: MEMBER(MemberID PK, Name); BORROW(BorrowID PK, MemberID FK, BookID FK)

	Q) A library stores MEMBER(MemberID, Name) and BORROW(BorrowID, MemberID, BookID). Find members whose number of borrowed books is above the average member borrowing count and rank them by borrowing count.
99	Full-Text Customer Support Relational Schema: SUPPORT_TICKET(TicketID PK, CustomerID FK, Subject, Description) Q) A company stores SUPPORT_TICKET(TicketID, CustomerID, Subject, Description). Design a full-text search that returns tickets matching both the terms 'payment' and 'refund', ordered by search relevance.
100	Full-Text Product Search Relational Schema: PRODUCT(ProductID PK, Name, Description, Keywords) Q) An e-commerce site stores PRODUCT(ProductID, Name, Description, Keywords). Design a full-text search for products related to the phrase 'wireless headphones' and return the most relevant matches first.